An Uncertainty-Aware Decision Support System for Hydraulic Pump Design Screening

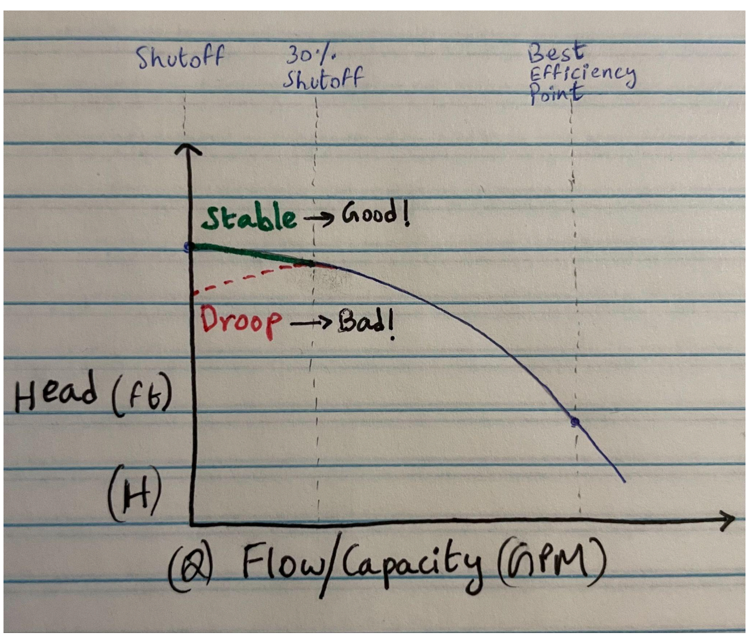
Design, deployment, and adoption of an uncertainty-aware ML decision support system during an industrial R&D co-op

By
Srujan Vaddiparthi
Former Data Science co-op (Jan - Dec ‘2025)
Hydraulics Engineering R&D
ITT Goulds Pumps

1. Problem

Engineers often need to decide whether a candidate design will behave reliably under real operating conditions **before** committing to expensive simulations, prototyping, or testing. These decisions are typically made under uncertainty, with limited historical data, strong safety constraints, and high costs associated with incorrect choices.

In industrial pump design, this challenge appears as the need to assess **hydraulic stability** across operating conditions before running computational fluid dynamics (CFD) simulations or lab tests, both of which are time- and resource-intensive. Industrial pumps are expected to operate reliably across a wide range of flow conditions. A key requirement, particularly in oil and gas applications, is hydraulic stability: the pump’s head-flow curve must rise smoothly and predictably, without exhibiting unstable or “drooping” behavior.



Stable vs Droop | Pump Performance (HQ) Curve

Selecting a pump design that satisfies this requirement is challenging. Engineers must decide whether a proposed design will exhibit a stable pump performance (Head-Flow) curve before committing to expensive and time-consuming validation steps such as Computational Fluid Dynamics (CFD) simulations or physical lab testing.

Each evaluation carries significant time, monetary, and opportunity cost, making early design decisions disproportionately impactful.

Today, this decision process relies on a combination of:
- ↳ Domain expertise and engineering intuition
- ↳ Historical designs and analogies
- ↳ CFD simulations
- ↳ Physical prototype testing

While these tools are effective, they are **slow, expensive and poorly suited for exploring large design spaces.** Running CFD or lab tests for every plausible design variant is impractical, yet exploring too few designs risks missing viable solutions.

A failed stability decision can have serious consequences: unstable designs may require costly redesigns, delay delivery schedules, or cause manufacturers to lose competitive bids. What is missing is a **fast, reliable way to screen candidate designs early,** while clearly communicating how confident the system is in each prediction.

Although this work was developed in the context of hydraulic pump design, the underlying challenges – limited data, high decision cost, interpretability requirements, and uncertainty-aware decision-making – are common across many applied engineering and data science domains.

2. Constraints and Design Requirements

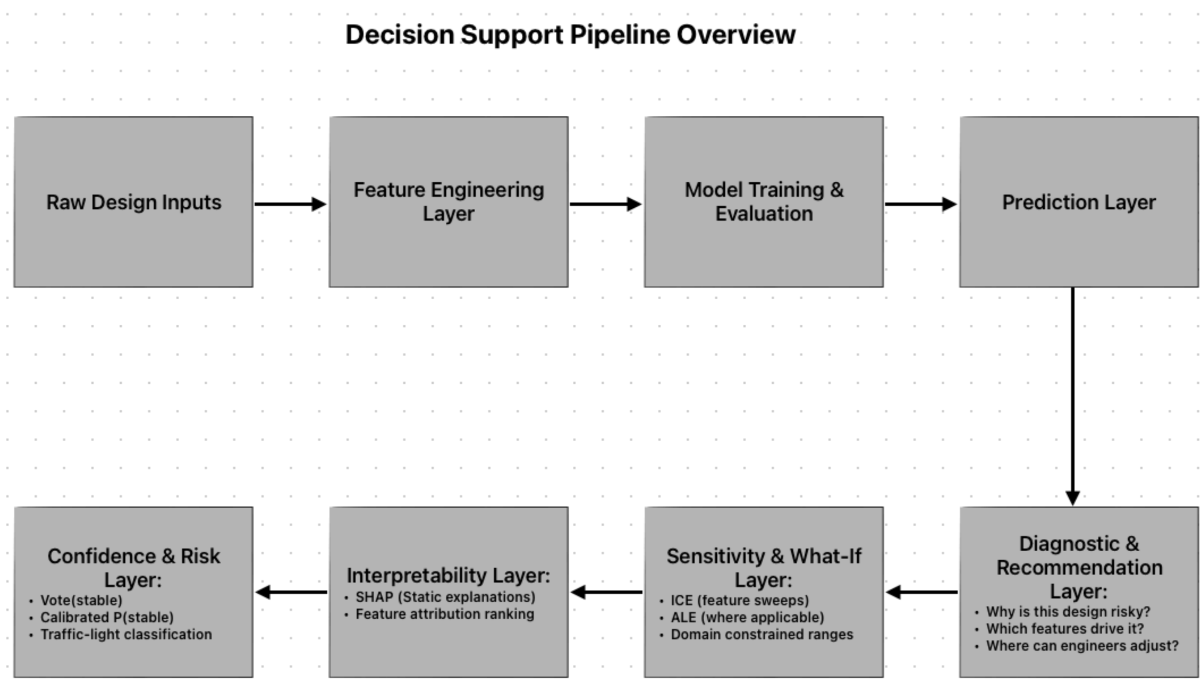
The decision-support system described in this document was developed under a set of practical constraints that significantly shaped the modeling, evaluation, and deployment strategy. These constraints are common in industrial engineering environments and require deliberate trade-offs beyond maximizing predictive accuracy alone.

- **Limited and Imbalanced Data:**
  The available dataset consisted of a relatively small number of historical pump designs, with a strong class imbalance between stable and unstable configurations. This limited the feasibility of data-hungry modeling approaches and increased the risk of overfitting. As a result, model robustness, stability across resampling, and consistent behavior under small perturbations were prioritized over single-fit performance.
- **High Cost of Incorrect Decisions:**
  Prediction errors carried asymmetric consequences.
  Incorrectly approving an unstable design could lead to costly redesigns, delayed delivery schedules, or failed bids, while overly conservative rejection of viable designs could result in missed opportunities. This imbalance required the system to explicitly communicate risk and uncertainty, rather than relying on point predictions alone.
- **Interpretability and Physical Plausibility:**
  Predictions needed to be explainable to practicing hydraulic engineers.

Model outputs were expected to align with physical intuition and known hydraulic relationships, enabling engineers to understand *why* a design was flagged as risky and *which parameters* were driving that assessment. Black-box models without interpretable explanations were therefore unsuitable, regardless of raw predictive performance.

- **Early-Stage Screening, Not Final Validation:**
  The system was designed to support early design screening, not to replace CFD simulations or physical testing. Its role was to help engineers rapidly narrow the design space and prioritize candidates for further analysis, while clearly conveying the confidence and limitations of each prediction.
- **Deployment and Operational Constraints:**
  The solution was required to operate in a local, offline engineering environment, without reliance on cloud infrastructure or frequent retraining. Models needed to be reproducible, stable across versions, and packaged into a form that could be used directly by engineers without machine learning expertise.

3. Approach:



The system was designed as a layered decision-support pipeline, with each component addressing a specific constraint identified in Section 2. Rather than optimizing a single model in isolation, the approach emphasizes robustness, uncertainty awareness, and interpretability under real-world engineering constraints.

3.1 Feature Design and Selection

Feature design combined statistical screening with domain expertise to ensure physical relevance and model stability.
Initial candidate features were evaluated using correlation analysis(pearson r), statistical tests (ANOVA/KW), and permutation-based importance measures, then reviewed with domain experts to eliminate redundant or non-physical inputs.

This process reduced the feature space from dozens of raw inputs to a compact interpretable subset per target, improving generalization in small-sample regimes while preserving meaningful hydraulic relationships. Feature selection was treated as a stability and trust-building step, not merely a dimensionality reduction exercise.

3.2 Ensemble Modeling for Robustness

Given the limited and imbalanced dataset, a single trained model was insufficiently reliable. Instead, the system employs an ensemble of models trained across multiple randomized seeds and stratified resampling schemes.

Ensemble agreement was used as a proxy for prediction reliability and robustness, reducing sensitivity to individual data points and mitigating overfitting. Model selection and tuning prioritized consistent behavior across resampling rather than peak single-fit accuracy, aligning model performance with real-world decision reliability. The ensemble framework was applied consistently across both regression-based stability metrics and downstream classification decisions, using agreement across models as a measure of robustness rather than relying on any single estimator.

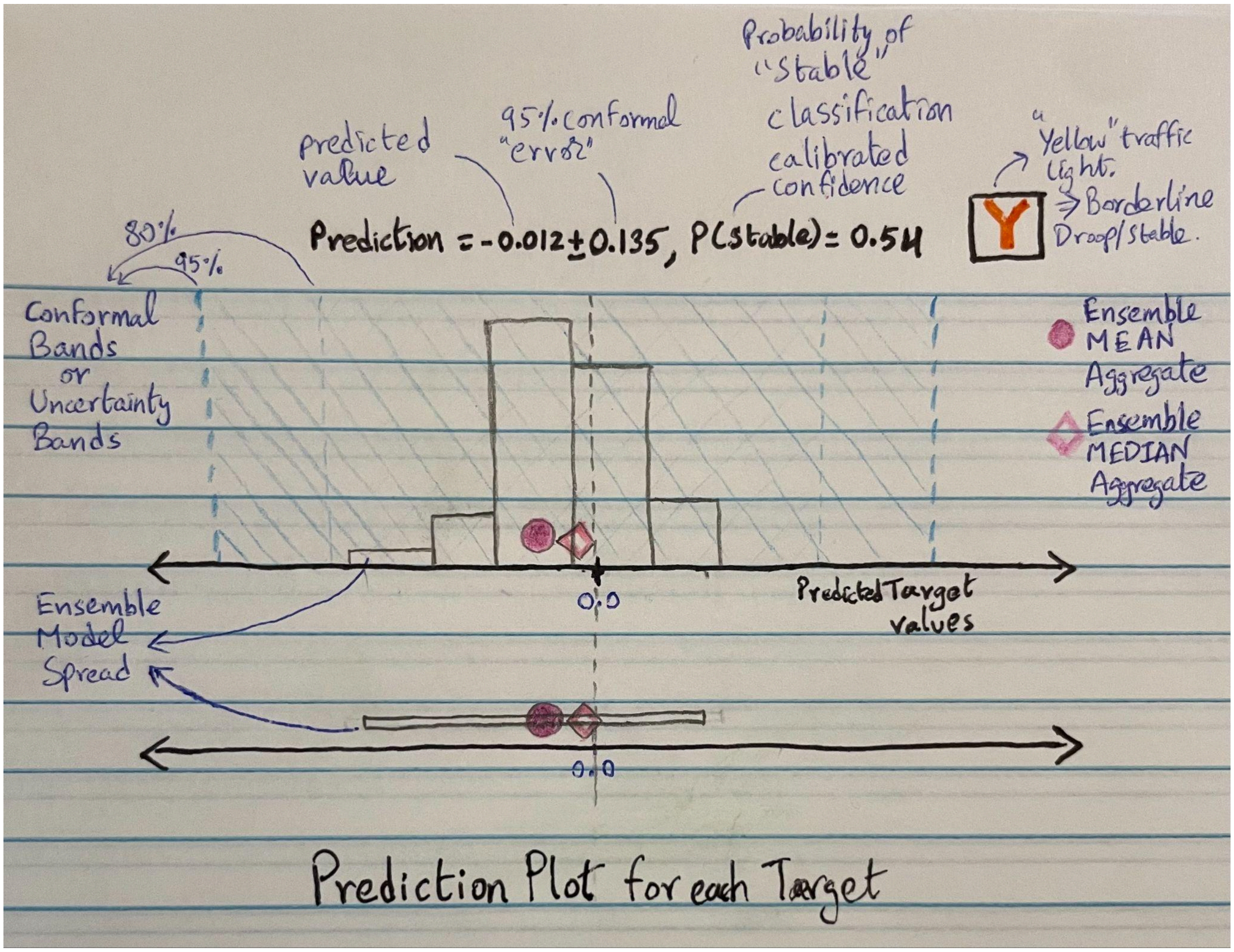
3.3 Stability-Aware Evaluation and Tuning

Hyperparameter tuning emphasized stability across validation splits rather than maximizing point estimates of performance.
Cross-validated metrics were evaluated not only by their mean performance but also by their variability, ensuring selected models behaved consistently under data perturbations.

This approach reduces the likelihood of brittle models that perform well on a single split but fail to generalize, which is particularly important in small-data, high-cost decision environments.

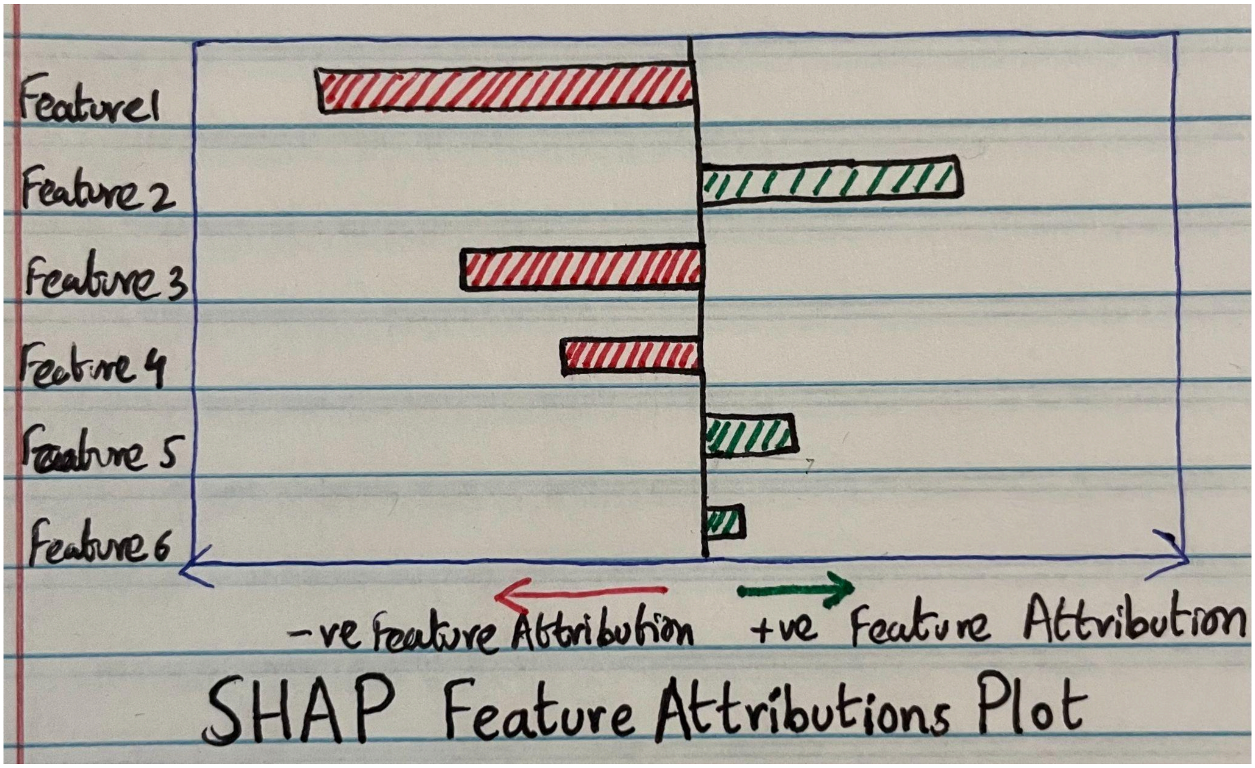
3.4 Uncertainty Quantification via Conformal Prediction

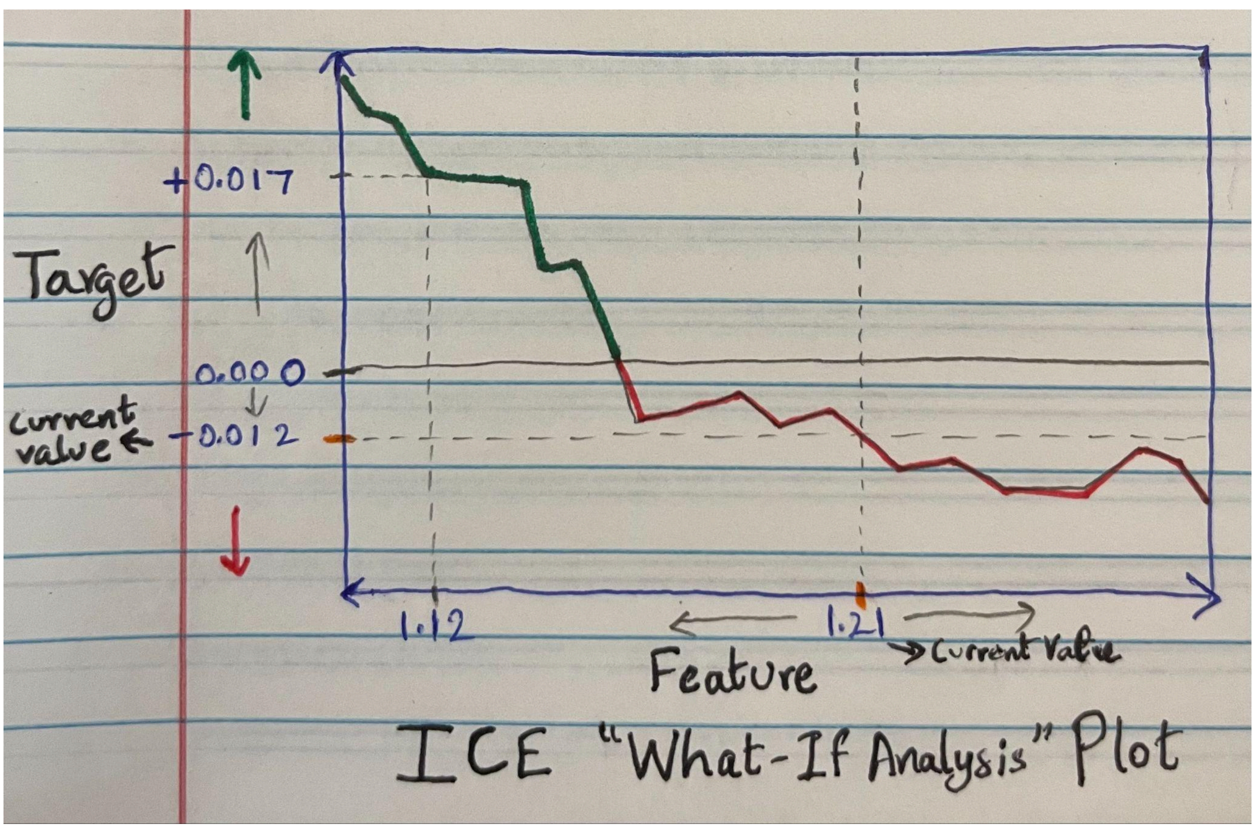
To support risk-aware decision-making, the system augments ensemble predictions with calibrated uncertainty estimates using a conformal prediction framework. Prediction intervals were derived from observed model errors and ensemble disagreement, producing coverage guarantees that remain valid in finite-sample settings.



Rather than reporting uncertainty as abstract statistical confidence, the resulting prediction intervals provide engineers with an intuitive sense of how reliable each prediction is, enabling informed trade-offs between risk and design exploration speed. Wider prediction intervals indicate lower agreement across ensemble members, while narrower intervals reflect higher model consensus and greater confidence in the prediction.

3.5 Explainability and What-If Analysis





Model interpretability was addressed through an engineer-facing explainability layer combining SHAP feature attributions with domain-constrained Individual Conditional Expectation (ICE) analysis. These tools allow engineers to:

→ Understand which parameters most strongly influence predictions,
→ Explore physically valid "what-if" scenarios,
→ And diagnose why a design is flagged as stable or unstable.

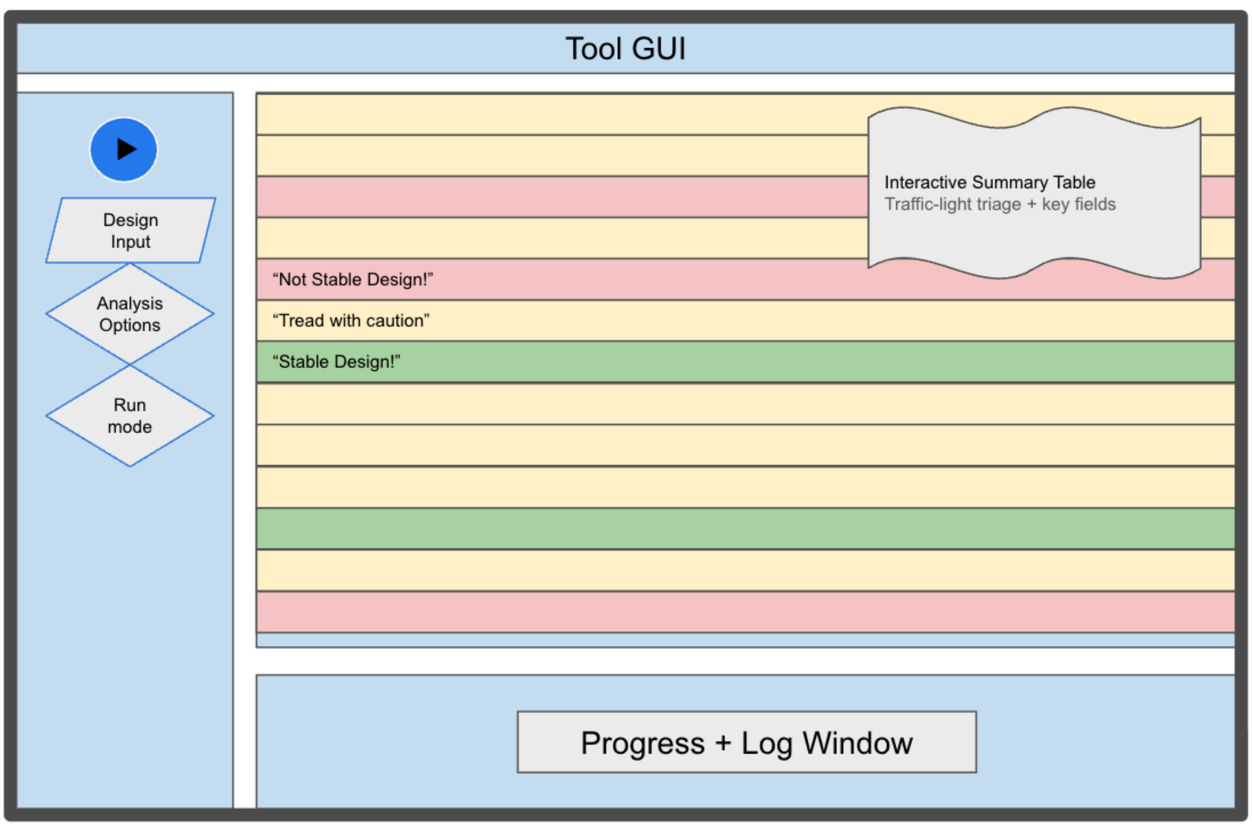
Explainability outputs were constrained to respect known physical bounds and coupled feature relationships, ensuring that model explanations remained actionable rather than purely statistical.

3.6 Integration into a Decision Support Tool

All modeling, uncertainty, and explainability components were integrated into a single desktop-based decision-support application. The tool presents predictions, uncertainty bands, and explanations together, allowing engineers to rapidly screen candidate designs and prioritize further analysis without interacting directly with machine learning internals.

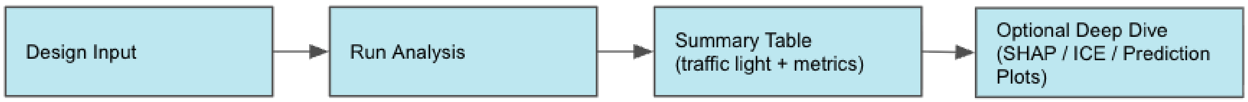
This end-to-end integration ensured that methodological advances translated into practical workflow improvements rather than remaining isolated analytical artifacts.

4. The Decision Support Tool:



All modeling, uncertainty estimation, and explainability components were integrated into a single locally deployable desktop decision-support application designed for use by hydraulic engineers during early-stage design screening.

Rather than exposing machine learning internals, the tool presents predictions, uncertainty, and explanations together in a unified interface, allowing engineers to rapidly assess candidate designs and prioritize further analysis.



4.1 User Inputs

Engineers provide a set of candidate design parameters derived from pump geometry and operating conditions.
Inputs are validated against known physical and feasibility constraints to prevent unrealistic or out-of-scope evaluations.

This allows the tool to be used safely without requiring ML expertise or manual data preprocessing.

4.2 Model Outputs and Risk Signals

For each candidate design, the tool presents:

→ A predicted stability-related performance value
→ A calibrated prediction interval representing uncertainty
→ A probability-based stability confidence score derived from ensemble agreement
→ A visual risk indicator highlighting borderline or high-risk designs

Prediction intervals expand when ensemble members disagree and contract when model consensus is high, giving engineers an intuitive sense of reliability rather than a single point estimate.

4.3 Interpretability and What-If Exploration

To support diagnostic reasoning, the tool includes an explainability and sensitivity analysis layer:

→ SHAP feature attributions indicate which parameters most strongly influenced each prediction
→ Domain-constrained ICE plots enable physically valid "what-if" exploration of design changes

These views allow engineers to understand why a design is flagged as stable or unstable and explore how targeted parameter adjustments could improve outcomes.

4.4 Workflow Integration and Adoption

The application was packaged as a standalone executable and deployed locally, requiring no cloud connectivity, retraining, or external dependencies.

The tool is actively used in design review discussions to:

→ screen large candidate sets in minutes,

→ focus CFD and lab testing on a small, high-leverage subset of designs,
→ and support faster, more confident early-stage decisions.

This end-to-end integration ensured that modeling advances translated directly into practical workflow improvements rather than remaining isolated analytical artifacts.

5. Impact and Engineering Relevance:

5.1 Practical Impact on Engineering Decision-making

The proposed system transforms predictive modeling outputs into a **risk-aware decision-support workflow** suitable for real-world engineering design contexts. Rather than providing point estimates in isolation, the tool presents predictions alongside calibrated uncertainty, ensemble agreement, and interpretable explanations, enabling engineers to assess both *expected performance* and *confidence in that performance* simultaneously.

This design reduces reliance on implicit trust in model outputs and instead promotes **informed judgment**, allowing engineers to distinguish between designs that are confidently stable, borderline, or high-risk before committing to downstream analysis or physical testing.

5.2 Robustness Under Data Scarcity and Design Variability

The framework was developed under conditions typical of industrial engineering settings: limited data availability, imbalanced outcome distributions, and high cost of incorrect decisions. By prioritizing ensemble agreement, stability-aware tuning, and conformal uncertainty estimation, the system remains reliable even in finite-sample regimes.

This robustness is particularly valuable during early-stage design exploration, where engineers must evaluate a large number of candidate designs with incomplete information and cannot afford brittle or overconfident predictions.

5.3 Interpretability as an Actionable Engineering Asset

Explainability within the system is not treated as a post hoc diagnostic, but as an **active decision aid**. Feature attributions and domain-constrained what-if analyses enable engineers to understand why a design is flagged as risky and which parameters can be adjusted to improve outcomes.

By constraining explanations to respect known physical bounds and coupled feature relationships, the system ensures that insights remain actionable rather than purely statistical, bridging the gap between machine learning outputs and engineering intuition.

5.4 Integration into Existing Engineering Workflows

The end-to-end integration of modeling, uncertainty quantification, and interpretability into a single desktop-based tool allows engineers to interact with advanced machine learning capabilities without requiring direct exposure to model internals. Summary-level decision signals support rapid triage, while optional drill-down enables deeper inspection when necessary.

This layered interaction design ensures that the system complements existing workflows rather than disrupting them, supporting faster iteration cycles and more confident design selection.

5.5 Generalizability Beyond the Present Domain

While developed within a specific engineering context, the methodology is **domain-agnostic** by construction. The principles of ensemble-based robustness, stability-aware evaluation, conformal uncertainty estimation, and constrained interpretability are applicable to a wide range of high-stakes decision environments, including other mechanical systems, manufacturing processes, and safety-critical design applications.

As such, the contribution extends beyond a single use case, offering a reusable blueprint for deploying trustworthy machine learning systems in real-world engineering settings.

5.6 Summary of Contributions

In summary, this work demonstrates that reliable engineering decision support requires more than predictive accuracy alone. By integrating robustness, uncertainty awareness, interpretability, and usability into a unified framework, the proposed system advances the practical adoption of machine learning in engineering design and risk assessment.

6. Conclusion:

This work presents the design, deployment, and adoption of an uncertainty-aware machine learning decision support system for early-stage hydraulic pump design screening. Developed under realistic industrial constraints, including limited and imbalanced data, high cost of incorrect decisions, interpretability requirements, and offline deployment, the system prioritizes robustness, uncertainty awareness, and usability over raw predictive accuracy. Rather than producing single point estimates, the framework integrates ensemble modeling, stability-aware evaluation, and conformal uncertainty quantification to communicate both predicted performance and confidence, enabling engineers to distinguish between confidently stable, borderline, and high-risk designs before committing to expensive downstream analysis.

Crucially, the system translates advanced modeling outputs into an engineer-facing workflow through a locally deployable decision support tool that combines calibrated prediction intervals, probability-based risk signals, and constrained explainability via SHAP and domain-aware what-if analysis. This layered interaction design supports rapid triage while allowing deeper diagnostic exploration when needed, without exposing machine learning internals. While developed in the context of hydraulic pump design, the methodology is broadly applicable to high-stakes engineering domains where data is scarce, decisions are costly, and trust in model outputs is essential. The work demonstrates that effective industrial ML systems must integrate uncertainty, interpretability, and workflow alignment as first-class design objectives, not afterthoughts.